WHITE BOOK

XPL201P

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1 Introduction

Modern industrial operations require efficiency, reliability, and safety in material handling. Automated Guided Vehicles (AGVs) address the challenges of high-volume, long-distance, and heavy-load transport in warehouses and factories.

1.1 Intended to use

The XPL201P is a heavy-duty Automated Guided Vehicle (AGV) designed for high-volume, high-speed, and heavy-load material handling in warehouses and complex industrial environments. It combines a robust mechanical structure with software-defined intelligence, modular hardware, and multi-sensor navigation to ensure precise, safe, and autonomous operation.

Fig 1.1: XPL201P

XPL201P supports continuous industrial operation, intelligent storage and retrieval, and collaborative fleet coordination, while enabling seamless integration with automated warehouse systems and enterprise software. The system is designed in compliance with ISO 3691-4, ensuring safe operation in industrial settings.

The AGV is intended exclusively for industrial material handling applications and must not be used for personnel transport.

Operating Modes

1. Automatic Mode (No Rider Allowed)

- Fully autonomous AGV operation
- Personnel transport is strictly prohibited
- All safety systems, including personnel detection and obstacle avoidance, are active
- Intended for automated production operation, continuous material transport, and fleet coordination

2. Manual Mode

- Operates like a conventional industrial pallet stacker
- Personnel detection systems are safely deactivated
- Intended only for short-distance movements, pallet pickup, and positioning
- Used in controlled environments under direct operator supervision

3. Maintenance Mode

- Intended exclusively for service, diagnostics, and maintenance activities
- Vehicle motion and safety functions are limited or disabled as required for maintenance
- Access restricted to trained and authorized personnel only

1.2 Key Functional Overview

Navigation & Safety

- Laser SLAM with optional reflector guidance for obstructed or dynamic environments
- 360° bottom safety LiDAR and safety PLC monitoring for safe collision prevention (Safety function PL d in accordance with the requirements of ISO 3691-4)
- Top camera and fork-tip LiDAR for obstacle detection
- Emergency stop switches

Material Handling

- Pallet recognition with automatic alignment adjustment
- Accurate load picking and unloading even with slight pallet misalignment
- Real-time storage location status (empty/full) for optimized scheduling

Operational Flexibility

- Support point-to-point, column-to-column, and collaborative tasks
- Compatible with multi-floor warehouse infrastructure
- Manual or automatic charging mode

System Integration

- Communicate with WMS (Warehouse Management System), MES (Manufacturing Execution System), ERP (Enterprise Resource Planning), and other enterprise systems.
- Provides real-time operational status, alarms, and task reporting

1.3 Primary Capabilities

The system provides autonomous storage management with automatic detection of storage availability, optimal unloading allocation, and precise pallet handling. It supports collaborative fleet operations, enabling multiple AGVs to perform both independent and synchronized transport tasks.

Task capabilities include standard operations such as point-to-point and column-based movements with intelligent task selection, as well as collaborative functions like docking, action, position coordination, and multi-vehicle binding.

Overall, the system enables flexible, efficient, and scalable multi-AGV operation for complex material handling environments.

1.4 Applications

XPL201P is designed for large-scale, high-throughput industrial environments, supporting long-distance transport, dense pallet storage, heavy-load handling, and continuous operation with high precision and reliability. It is well suited for warehouses and production facilities requiring efficient material flow and integration with automated infrastructure such as elevators, conveyors, doors, palletizers, and robotic arms.

The system features a warehouse-free storage location management approach, enabling autonomous perception, decision-making, and automatic forking without fixed storage definitions. This improves operational efficiency while addressing labor shortages, high turnover, unclear storage zoning, and inefficient stacking or sorting practices.

At the warehouse-area level, XPL201P provides intelligent area-based control, supporting flexible unloading and picking strategies, bidirectional traffic, real-time empty/full status monitoring, and selective or full-area clearing and filling. These functions enhance storage flexibility, space utilization, and unloading efficiency.

Implementation functions such as route teaching, visualized and editable layouts, customizable storage areas, and task definitions enable rapid deployment and site-specific configuration, ensuring adaptability across different operational scenarios.

1.5 Compliance and Safety Standards

The XPL201P is designed and developed in accordance with relevant international and European standards to ensure safe operation, reliable performance, and compliance with applicable regulatory requirements for automated material handling systems.

- **Machinery Directive 2006/42/EC**

  The XPL201P complies with the essential health and safety requirements of the European Machinery Directive 2006/42/EC, forming the basis for CE conformity and safe integration into industrial environments.

- **ISO 3691-4**
  Industrial trucks — Safety requirements and verification — Part 4: Driverless industrial trucks and their systems

  Referenced under the Safety System section:
  "The safety system mainly consists of a safety PLC controller and various safety sensors to form a safety monitoring system to meet the requirements of ISO 3691-4 standard. The XPL201P has been designed, verified, and validated in accordance with the requirements of ISO 3691-4."

- **EMC Directive 2014/30/EU (Electromagnetic Compatibility)**
  The system fulfills the requirements of the EMC Directive 2014/30/EU, ensuring that the XPL201P does not generate electromagnetic disturbances that could affect other equipment and is itself resistant to external electromagnetic influences, guaranteeing reliable operation in industrial environments.

1.6 Truck parameters

Table 1.1:Basic Parameters

Parameter	Symbol	Unit	Value
Manufacturer			EP
Model name			XPL201P
Model Designation			Automated guided vehicle
Drive			Electric
Load Capacity		kg	2000
Service Weight		kg	625
Navigation			Slam/Reflector
Communication			Wi-Fi/5G
Positioning Accuracy		mm	± 10
Indoor/Outdoor			Indoor

Battery Parameters

Parameter	Symbol	Unit	Value
Battery Type			Li-ion Battery
Battery Voltage/ Nominal Capacity		V/Ah	24/205
Usage Time		Hours	6-8

Dimensions

Parameter	Symbol	Unit	Value
Load Centre Distance	c	mm	600
Load Length	x	mm	980
Wheelbase	y	mm	1430
Height with Fork Lowered	h13	mm	90
Lifting Height	h3	mm	130

Other Parameters

Parameter	Symbol	Unit	Value
Travel speed		m/s	up to 2.5 m/s (adjustable per load and safety zones
Max. Gradeability		%	8/10
Maximum Floor Level Devi-ation		mm	≤8
Turning Radius	Wa	mm	1785

Aisle Requirements

Parameter	Symbol	Unit	Value
Aisle width for pallets 1000×1200	Ast	mm	1800
Aisle width for pallets 1000×1200	Ast	mm	2500
Aisle width for one-sided	Ast	mm	3600/3000

Auto-Charge

Parameter	Symbol	Unit	Value
Communication			Wi-Fi
Charging Power			24V/100A

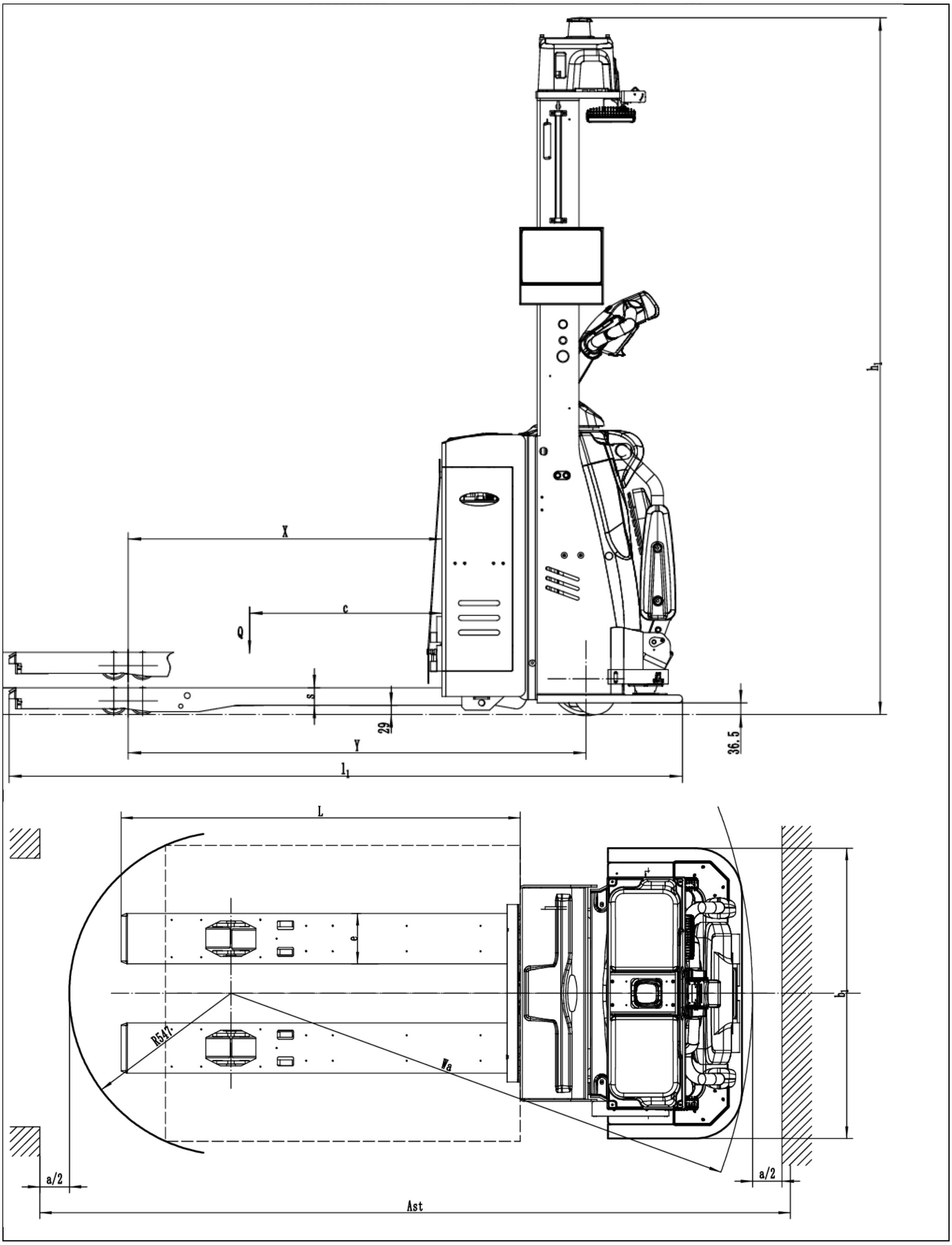

Fig 1.2: Dimensions

1.7 Key Benefits

- Labor efficiency through autonomous operation
- Consistent and predictable workflows
- Enhanced safety with multi-layer obstacle detection
- Real-time integration with WMS, MES, ERP

2 Main truck components and description

2.1 Physical Layout

These diagrams illustrate the physical layout of major components including navigation sensors, safety devices, control units, drive systems, and material handling mechanisms, providing a clear reference for installation, maintenance, and troubleshooting.

Fig 2.1: Right side Components

Fig 2.2: Left Side Components

Fig 2.3: Back/Fork side components

2.2 Navigation System

Laser SLAM (Simultaneous Localization and Mapping) technology enables the generation of precise 3D digital representations of environments. By utilizing LiDAR sensors, the system continuously scans its surroundings while simultaneously determining its own position within the mapped space. The result is a highly detailed 3D point cloud that captures spatial structures such as walls, racks, and obstacles.

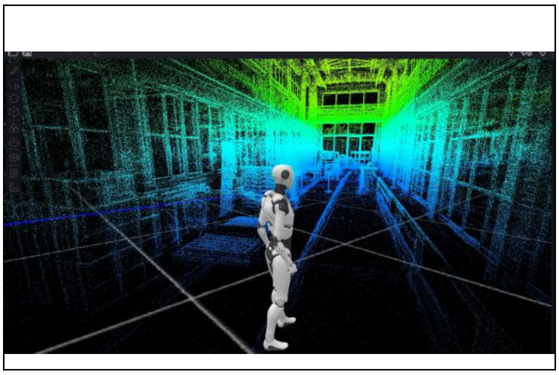

Fig 2.4: 3D-point cloud picture

For navigation, two complementary methods are used: natural Laser SLAM guidance, which enables flexible operation without infrastructure, and reflector-based guidance, which enhances positioning accuracy in complex or feature-poor areas. Together, these approaches ensure reliable and precise navigation across a wide range of warehouse environments.

As illustrated in Fig. 2.4, the generated point cloud provides a comprehensive digital representation of the warehouse environment, allowing the system to perceive depth, geometry, and navigable space with high accuracy. This digital map forms the foundation for reliable and autonomous navigation.

Navigation Methods

1. Laser SLAM Guidance

- Purpose: Enables autonomous positioning and route planning in natural environments.
- Working Principle: LiDAR scans the surroundings to build a real-time map; SLAM algorithms compute the vehicle's position relative to the map.

- Benefits: Accurate navigation in large, complex warehouses without physical markers; suitable for general, open, or dynamic areas.

2. Reflector Guidance

- Purpose: Enhances positioning accuracy in densely obstructed or cluttered environments.
- Working Principle: Reflective markers (e.g., reflective tape) are installed at fixed locations within the environment. These markers generate strong and easily identifiable LiDAR reflections, which act as stable reference points within the point cloud. By incorporating these references into the SLAM process, the system improves localization accuracy, particularly in areas where natural features are insufficient.
- Benefits: Reliable navigation in areas with dense stacks, shelving, and charging stations, where natural feature detection may be insufficient.
- Default Method: Natural SLAM navigation is used as the standard approach, while reflector guidance is applied selectively where enhanced positioning accuracy is required.

2.3 Upper 3D Camera

- Purpose: Supports obstacle detection by identifying frontal and low-hanging objects, assisting the vehicle in navigating through dynamic environments.
- Working Principle: A depth camera captures real-time 3D point cloud data of the area in front of the vehicle. This data is processed by an onboard industrial computer to estimate distances and detect potential obstacles. Based on this analysis, the system provides input to the vehicle control system, enabling timely speed adjustments or controlled deceleration via torque control.
- Benefits: Enhances operational awareness by assisting in the detection of pedestrians, forklifts, and suspended obstacles. This contributes to smoother navigation and reduces the likelihood of disruptions such as minor collisions or traffic jams in busy warehouse environments.
- Implementation: The overhead-mounted camera continuously scans the ground ahead and nearby obstacles, improving overall environmental perception in real time.

2.4 Bottom 360° Safety LiDAR

- Purpose: Provides comprehensive ground-level coverage to detect surrounding obstacles and support safe vehicle operation. Safety component verified to achieve Performance Level d in accordance with EN ISO 13849-1.
- Working Principle: The system utilizes three LiDAR sensors (mounted at the front and under the fenders) to achieve 360° environmental scanning. The detected data is processed in real time to generate a safety field around the vehicle, which is divided into four configurable zones. In the outer zones, the system issues warnings and progressively reduces vehicle speed as obstacles are detected closer to the vehicle. The innermost zone (red zone) represents the final safety boundary. When an object is detected within this zone, the system triggers an immediate and controlled stop of the machine.
- Benefits: Enables reliable obstacle detection and controlled vehicle response in tight and dynamic warehouse environments. The multi-zone concept allows for gradual deceleration, improving operational smoothness while ensuring a defined safety stop when required. In the presence of stationary obstacles, the system typically detects objects within the outer warning zones at an early stage. This allows the vehicle to decelerate progressively, preventing entry into the final safety (red) zone. As a result, unnecessary emergency stops are avoided, leading to smoother operation, reduced mechanical stress, and minimized wear on braking components.

2.5 Fork Tip LiDAR and Camera

- Purpose: Enables precise pallet detection and handling while supporting safe operation during reversing and load engagement.
- Working Principle: A combination of LiDAR and depth camera sensors mounted at the fork tips continuously monitoring the area in the fork direction. The system detects pallets and low-profile obstacles close to the ground with high accuracy. During pallet handling operations, the system can selectively mute person detection in the fork direction once a pallet is reliably identified. This controlled muting allows the vehicle to approach and engage the load without unnecessary interruptions. To maintain safety during this mode, the vehicle speed is automatically limited to a maximum of 0.3 m/s. In all other situations, obstacle detection remains active, and the system triggers deceleration or braking when objects are detected.
- Benefits: Ensures accurate and efficient pallet handling while maintaining a high level of operational safety. The intelligent muting function enables smooth load engagement without false stops, while the reduced- speed during this mode minimizes risk. Additionally, the system helps prevent collisions and damage to pedestrians, equipment, and goods during reversing and forking operations.

2.6 Safety PLC

- Purpose: Central monitoring and control for all safety mechanisms.
- Working Principle: Processes inputs from emergency stops, LiDARs, encoders, and proximity switches; controls speed limits, direction checks, and emergency stops.
- Benefits: ISO 3691-4 and EN ISO 13849-1/-2 compliance, protects personnel and equipment. Safety component verified to achieve Performance Level e in accordance with EN ISO 13849-1.

2.7 Emergency Stop Switches

- Purpose: Immediate manual shutdown in emergency situations
- Working Principle: Dual independent signals to Safety PLC; disconnects main power contactors for emergency braking.
- Benefits: Redundant safety layer, Safety component verified to achieve Performance Level d in accordance with EN ISO 13849-1.

2.8 Safety Encoder, Rotary Proximity, Load Switch

- Purpose: Monitors speed, direction, and load status.
- Working Principle: Sends signals to PLC; triggers braking or speed limit enforcement under unsafe conditions.
- Benefits: Maintains safe operation under varying loads and turns. Safety component verified in accordance with the requirements of ISO 3691-4.

2.9 Acoustic and Light warning Systems

- Purpose: Enhances operational safety by alerting surrounding personnel to vehicle movement, abnormal conditions, and operational status in real time.
- Working Principle: Integrated acoustic alerts, flashing lights, and turn signals indicate task status, faults, warnings, or directional movement.
- Functions:
  - Acoustic alerts for task progress, faults, and warnings
  - Red warning light for forward motion

  - Turn signals for directional changes
  - Automatic alarms triggered by obstacles, communication errors, positioning failures, low battery, or manual operation zones
- Benefits: Ensures personnel safety, situational awareness, and prompt response to operational issues.

2.10 In-Car Interaction

The vehicle provides intuitive onboard human–machine interaction interfaces by touch screen or multi-key button interface. It allows operators to monitor status, perform diagnostics, and execute manual operations when required.

Operator input during automatic mode is implemented in accordance with EN ISO 3691-4 (4.9.2.2). The vehicle stops at defined positions and allows manual actions under controlled conditions, including hold-to-run operation, no vehicle movement, and a deliberate restart of the automatic sequence.

2.11 Task Triggering and Equipment Interface

Tasks can be triggered using multiple calling devices:

- Button box
- PAD
- Mobile phone
- Computer
- Aerial camera

Purpose: Enables flexible task initiation from different operational positions.

The XPL201P can interact with various fixed and automated systems:

- Elevators (multi-floor, single or double door)
- Lifts
- Automatic doors
- Air shower rooms
- Robotic arms
- Palletizers
- Conveyor lines

Purpose: It allows seamless integration into automated production and logistics systems.

2.12 System Integration

The system supports standard enterprise-level integration with platforms such as WMS (Warehouse Management System), MES (Manufacturing Execution System), and ERP (Enterprise Resource Planning) systems.

Purpose: Enables real-time task dispatching, seamless data exchange, and enhanced operational transparency across intralogistics and production processes.

Cybersecurity: The system is designed with secure communication interfaces and access control mechanisms to ensure reliable and protected data exchange with connected enterprise systems, supporting safe integration within IT infrastructures.

3 Product function definition

The XPL201P performs fully autonomous pallet transport and storage/retrieval operations inside defined warehouse zones. All movements and handling actions are executed according to tasks received from a fleet management system or triggered manually under controlled conditions. The vehicle integrates navigation, motion control, obstacle detection, pallet alignment, and independent safety supervision to achieve repeatable positioning and safe operation in compliance with ISO 3691-4.

3.1 Navigation system logic description

The XPL201P navigation system enables precise autonomous movement in complex warehouse environments. It combines laser-based SLAM for natural-feature navigation (the default and primary method) with an optional reflector-based enhancement for specific challenging zones. The architecture integrates real-time localization, route planning and path following while working in parallel with the obstacle detection and safety systems.

3.1.1 Navigation laser (standard)

The default navigation method is natural-environment SLAM. A single navigation laser scanner continuously scans the surroundings and produces a real-time point cloud of the environment. Onboard software matches this live data against a pre-built digital map of the warehouse, allowing the vehicle to always determine its exact position and orientation.

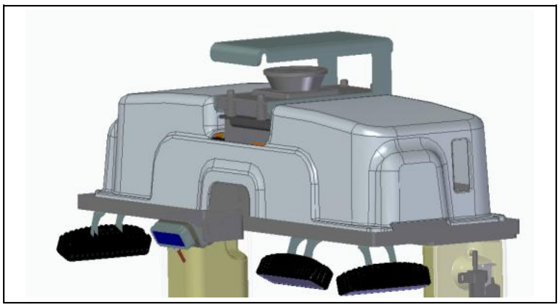
Fig 3.1: Navigation Laser

This method relies solely on existing environmental features (racking, columns, walls, machinery etc.) and requires no additional infrastructure in standard operation. It performs well in typical warehouse layouts with adequate natural variation.

In zones with repetitive, sparse or highly dynamic features — such as dense racking, charging stations or areas with similar columns — an optional reflector enhancement can be activated. Narrow reflective tape strips are placed at key points. The same navigation laser detects these high-reflectivity landmarks and integrates them into the SLAM process, improving localization robustness without changing the core natural method.

Positioning accuracy remains ±10 mm under normal conditions (clean floor, stable lighting, no extreme dynamics). The system continuously verifies path adherence and travel direction. If localization confidence drops below threshold, the vehicle slows, attempts recovery using recent data, and stops if needed.

<u>Key functions performed by the navigation system include:</u>

- Continuous real-time localization within the mapped environment
- Map matching between live scan data and the stored map
- Route planning and precise trajectory following

- Verification of forward direction and path adherence
- Optional feature enhancement through reflective markers in defined zones

3.2 Person and obstacle detection System

The XPL201P is equipped with a multi-sensor detection system that supports safe and efficient autonomous operation. It provides real-time awareness of obstacles during travel and assists with precise pallet handling during approach, pickup and reversing. The system combines several sensors that operate in parallel and supply processed data to the onboard industrial computer for appropriate speed- and direction-dependent reactions.

Detection zones and responses adapt automatically to the current vehicle speed, travel direction and maneuver type (detailed zone definitions and speed-dependent logic are provided in Chapter 4). When an obstacle is detected in a defined area, the system issues progressive warnings, reduces speed or applies controlled braking as required. These detection functions improve operational smoothness and situational awareness but are separate from the independent, certified safety supervision layer (see 3.2).

3.2.1 Upper 3D camera

An upper-mounted 3D depth camera continuously monitors the forward area and close-to-ground space ahead of the vehicle. It detects pedestrians, other vehicles, suspended loads and low-level obstacles in the travel path.

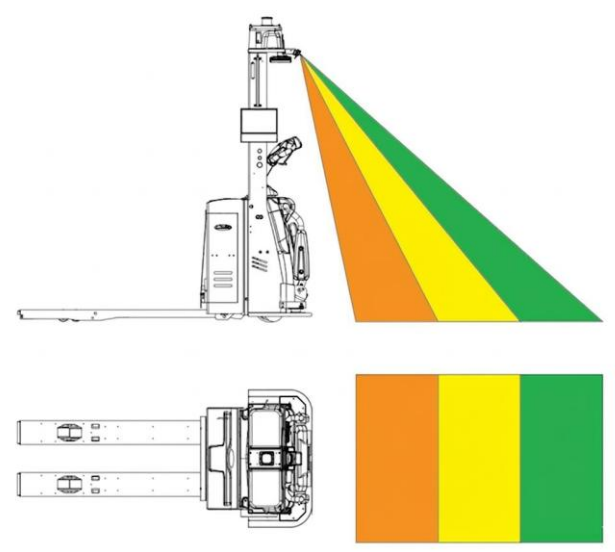

Fig 3.2: 3D Camera

The camera generates point-cloud data that enables the system to identify objects at different distances and initiate the following operational responses:

- When an obstacle enters the warning zone (yellow), an audible alert and visual indication are activated, and the vehicle significantly reduces speed.
- When an obstacle enters the critical zone (orange), the vehicle comes to a full stop using torque braking commanded through the traction controller.

This camera function assists normal driving and handling operations by providing additional environmental information.

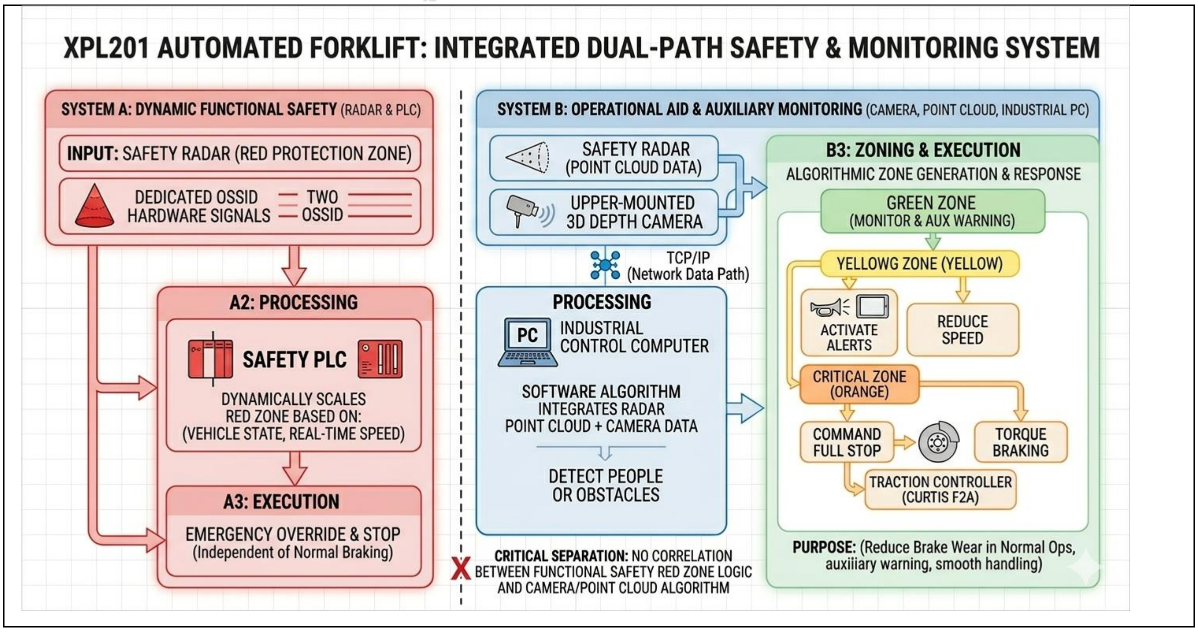

Fig 3.3: Logical Diagram

3.2.2 360° PLD LiDAR

Three LiDARs (positioned at the front and two sides of the fenders) provide continuous 360° ground-level coverage around the vehicle perimeter. These sensors detect low-level obstacles such as pallets on the floor, protruding legs, pedestrians' feet, small items or other ground hazards that could interfere with safe movement.

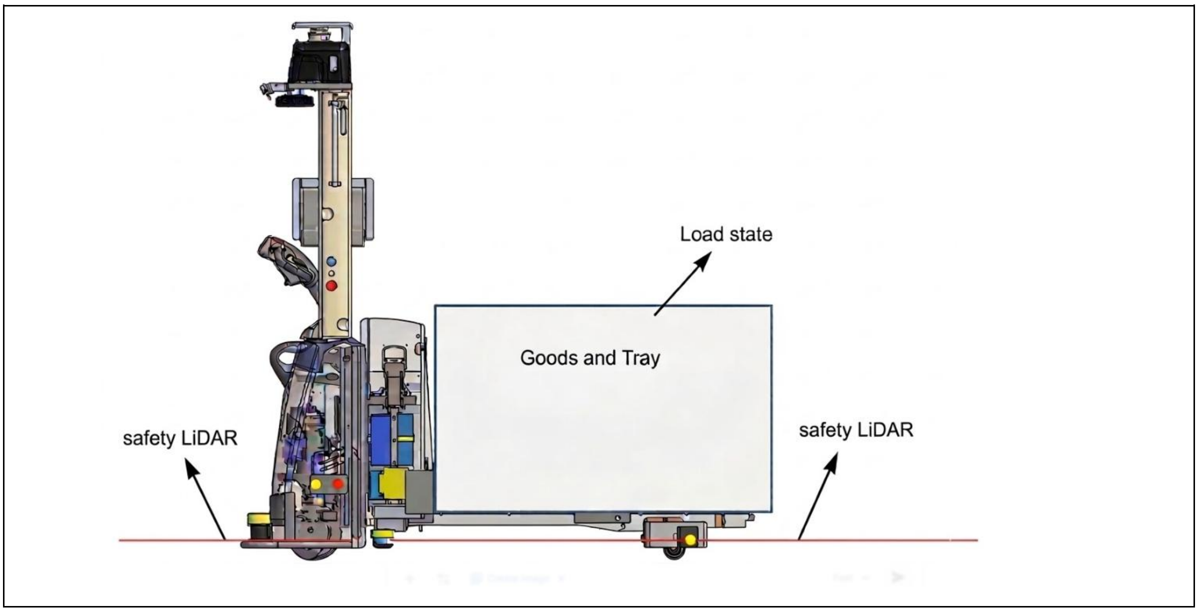

Fig 3.4: Bottom safety LiDAR

The LiDARs generate real-time point-cloud data that is processed to define multiple concentric protection zones.

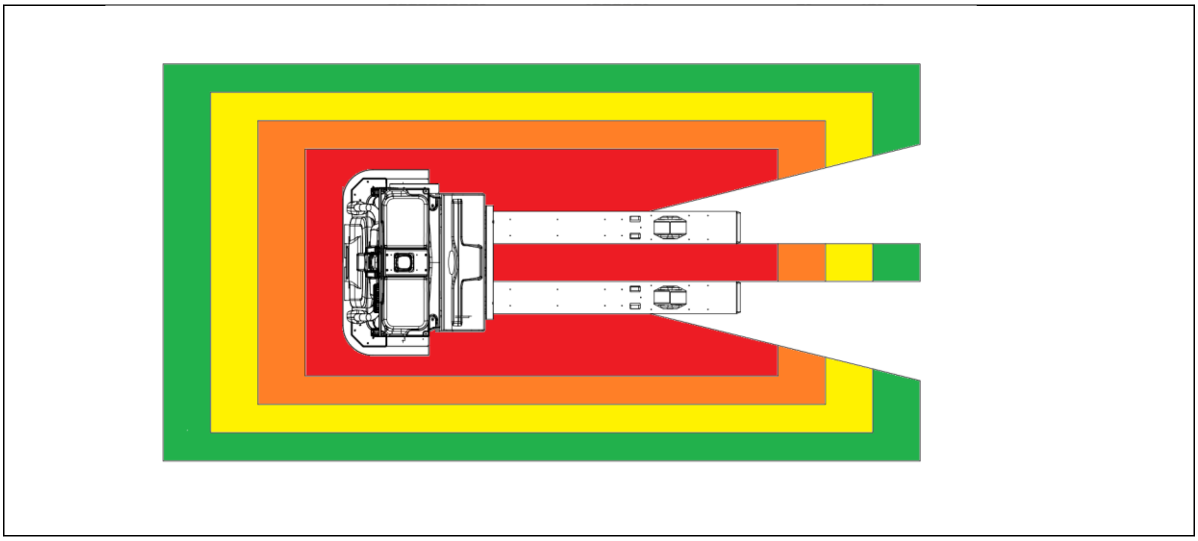

Fig 3.5: LiDAR Zones

These zones adjust automatically based on the vehicle's speed, direction of travel (forward, reverse or turning) and maneuver type. The response logic is progressive:

- Warning area (green): Audible and visual alert issued; vehicle applies moderate deceleration to allow time for the obstacle to clear or for the AGV to slow safely.
- Deceleration area (yellow): Stronger controlled deceleration activated via torque braking through the traction controller.
- Parking/stop area (orange): Vehicle comes to a full controlled stop using torque braking.
- Red (safety zone): This innermost zone is monitored by the independent Safety PLC (Category 3 PL d rated LiDARs according to EN ISO 13849-1 feeding into PL e/d safety chain). Entry into the red zone triggers immediate opening of dual contactors, cutting power to both the traction controller and the steering controller. The spring-applied electromagnetic brake (EB) engages for a fail-safe stop. Resumption of motion requires obstacle clearance, manual reset, and a 3-second delay with audible and visual warnings.

3.2.3 Fork tip obstacle avoidance LiDAR

A dedicated LiDAR is mounted at the fork tips to monitor the area behind and to the sides during reversing maneuvers. Its primary function is to detect low-level obstacles (such as floor debris, protruding pallet legs, small items or people close to ground level) in the reversing direction. The detection coverage is wider than the vehicle's maximum load width to provide effective rear protection.

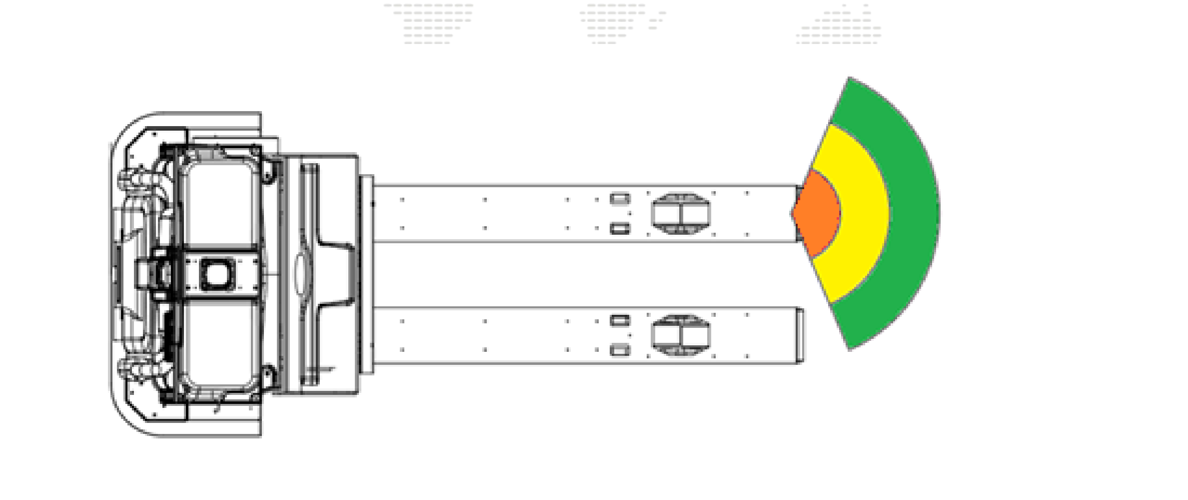

Fig 3.6: Fork tip obstacle liadar

The LiDAR scans a horizontal plane close to the ground and generates real-time point-cloud data (as illustrated in the top-view detection zones: green warning, yellow deceleration, orange stop areas). Data is transmitted via RJ45 (TCP/IP) to the industrial control computer, which processes distances and defines adaptive zones based on current speed and direction.

During autonomous reversing operation:

- When an obstacle enters the warning area (green), the industrial computer sends signals via CAN to the acoustic module and I/O control board, triggering audible and visual alerts.
- When an obstacle enters the deceleration area (yellow), the computer commands moderate controlled deceleration to the traction controller, using torque braking.
- When an obstacle enters the stop area (orange), the computer commands a full stop to the traction controller, applying torque braking for a controlled halt.

This LiDAR assists safe reversing and handling in tight spaces but operates as a non-safety-related (operational) system. It is separate from the certified bottom 360° safety LiDARs and Safety PLC layer.

3.2.4 Fork-tip Camera Obstacle Avoidance

A 3D depth camera mounted at the fork tips supports two main functions during autonomous operation: obstacle detection in the reversing direction and precise pallet recognition for accurate fork insertion

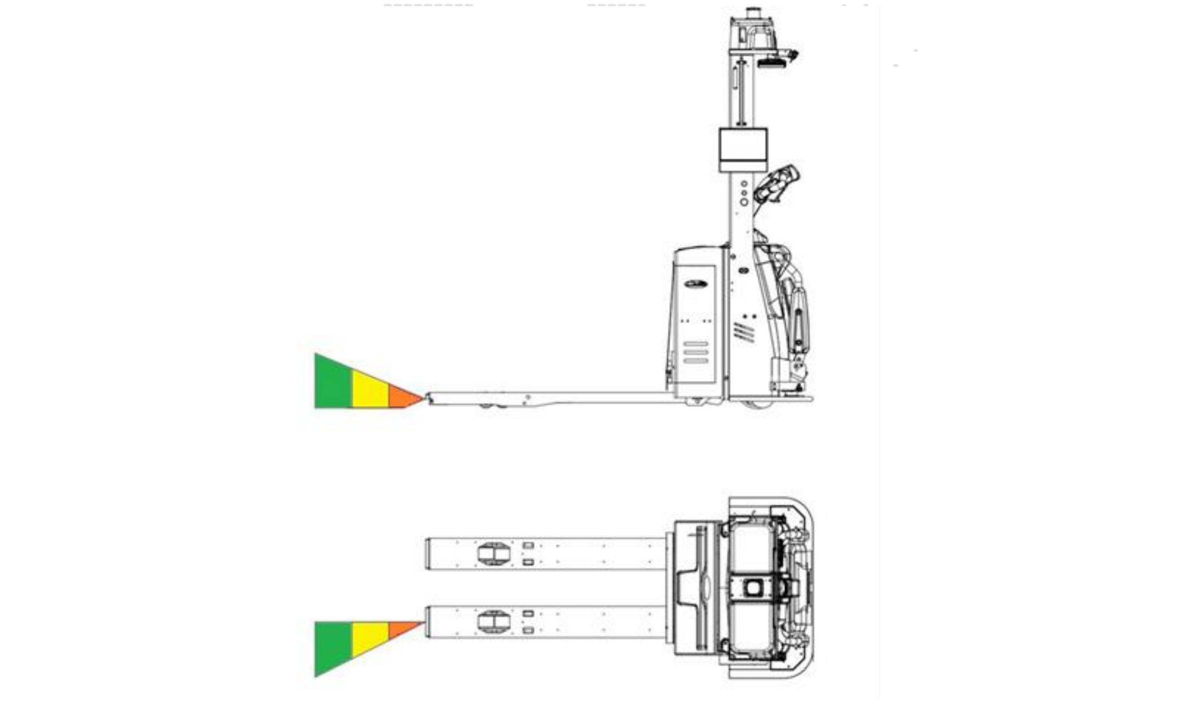

Fig 3.7: Fork-tip Camera Obstacle Avoidance

The camera monitors the area behind and around the forks (coverage wider than the vehicle's maximum load width) and generates real-time depth-based point-cloud data. This data is transmitted via USB 3.0 to the industrial control computer for processing.

Obstacle detection during reversing: The camera identifies low-level hazards (e.g., floor debris, protruding pallet legs, small items, or people close to ground level). Detection zones adapt to speed and direction, as shown in the top-view illustrations (green warning, yellow deceleration, orange stop areas). Response logic is progressive:

- Obstacle in warning area (green) → audible and visual alerts via acoustic module and I/O board (CAN communication).
- Obstacle in deceleration area (yellow) → controlled torque braking command to the traction controller.
- Obstacle in stop area (orange) → full controlled stop command to controller using torque braking.

Pallet recognition and fork insertion During approach, the camera scans the pallet front face and entry pockets of standard three-stringer pallets. It detects lateral offset, angular deviation, and pocket positions, then automatically calculates and applies small lateral and rotational corrections to align the forks accurately before insertion. This improves pickup reliability even when pallets are placed with reasonable inaccuracy.

The system remains effective with partial shrink-wrap or foil coverage (up to ~40% of the pallet surface). Depth-based geometry analysis (rather than relying on surface texture or color) allows reliable identification of structural features (openings, stringers, overall shape) despite reflective or occluded areas. In cases of extreme foil coverage or heavy distortion, performance may vary, but the camera's 3D nature provides good robustness compared to 2D vision systems.

This fork-tip camera function enhances both reversing awareness and pallet handling precision but operates as a non-safety-related (operational/assistive) system. It is separate from the certified bottom 360° safety LiDARs and Safety PLC supervision layer.

3.3 Safety control system

XPL201P is equipped with an independent safety control system designed to meet the requirements of ISO 3691-4. This system operates separately from the navigation and task execution software and has the highest authority to interrupt motion and force a fail-safe stop when necessary.

3.3.1 Safety PLC

The safety PLC is the central monitoring unit of the safety system. It continuously supervises critical vehicle parameters and enforces safe operation according to ISO 3691-4 performance level requirements.

Its main functions are:

- Safety speed monitoring detects overspeed, stall, wrong direction, or speed limit violations (forward/reverse, loaded/unloaded) using the safety encoder.
- Safety direction monitoring recognizes left/right turn direction via rotary proximity switches and switches LiDAR protection zones automatically.
- Safety LiDAR monitoring: independently supervises the bottom 360° safety LiDAR protection fields.
- Safety emergency stop: processes signals from all emergency stop switches.
- Additional safety signal monitoring: supervises load switch status, bumper activation, contactor feedback, and other I/O signals.

Fig 3.10: Safety PLC

3.3.2 Emergency stop switch

The XPL201P is equipped with three emergency stop switches to ensure rapid access and compliance with ISO 3691-4 and ISO 13850:

- Two side-mounted emergency stop switches (one on each side of the vehicle, positioned for easy reach from the ground or during maintenance).
- One central/main emergency stop switch (red button located on the mast/column, clearly visible and accessible to the operator during manual mode).

Each switch provides two independents normally closed (NC) signals connected directly to the safety PLC. The PLC continuously monitors the status of all three switches in real time.

When any one of the emergency stop switches is pressed (whether in automatic or manual mode), the safety PLC immediately detects the signal disconnection. It then:

- Opens the two main contactors.
- Cuts power to the controllers.
- Engages the spring-applied electromagnetic brake (EB) for a fail-safe emergency stop.

Fig 3.11: Emergency Stop Switch

The system uses closed-loop verification: the PLC monitors the normally closed auxiliary contacts of the main contactors to confirm that power has been successfully removed.

To resume operation after an emergency stop:

- The obstacle or fault condition must be cleared.
- The pressed emergency stop switch(es) must be released (twisted and pulled back to the normal position).
- The operator presses the reset button.
- After a mandatory 3-second delay accompanied by audible and visual warnings, the vehicle is permitted to restart and continue the task.

4 Operating Conditions

This chapter defines the permissible environmental conditions, ground/floor requirements, personnel safety rules, and typical usage scenarios for the XPL201P automated pallet truck. Compliance ensures stable navigation (±10 mm accuracy), reliable performance, and safe operation in accordance with ISO 3691-4.

4.1 Operating requirements

4.1.1 Power requirements

- Nominal voltage: 230 V AC, single-phase, 50 Hz
- Permissible fluctuation: ±10%
- Protective grounding (earth) required in the power distribution system for electrical safety
- No specific protection for connection plates mentioned; standard industrial connectors are used (ensure proper strain relief and IP rating per site conditions)

4.1.2 Equipment operating environment requirements

The XPL201P is designed for indoor industrial environments with the following limits:

- Operating temperature: -10 °C to +40 °C
- Charging temperature: 0 °C to +40 °C (battery type dependent; consult manual for exact limits)
- Relative humidity: up to 90 % RH, non-condensing
- Avoid direct sunlight on laser navigation sensors
- Avoid large reflective surfaces (glass, mirrors, polished metal) or pure black/absorbing areas in the laser scan field; if unavoidable, frost glass or shield surfaces
- No corrosive gases, explosive atmospheres, or harmful liquids
- Low dust and metal particle concentration
- No continuous vibration or excessive mechanical shock

4.1.3 Floor requirements

The driving surface must be smooth, clean, and stable to support reliable laser SLAM navigation and mechanical performance. Requirements follow general AGV guidelines (EN 15620, VDI 2510, ISO 3691-4 expectations):

Surface Undulation

Undulation is defined as the height difference between the highest and lowest points within a reference area.

- Maximum allowable undulation within 1 m²: ≤ 3 mm
- The surface must be:
  - Clean
  - Free of loose particles and debris
  - Non-slip

Excessive undulation may cause vibration, reduced navigation accuracy, and increased wear on the wheel and drive system. Maintaining a smooth surface ensures stable motion and precise positioning.

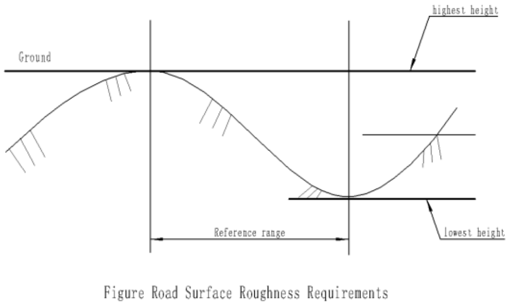

Fig 4.1. Road surface smoothness requirements

Road Surface Slope

- Road surface slope (H/L) is defined as the ratio between height difference (H) and route length (L) over a minimum distance of 100 mm.
- Maximum allowable slope for travel:8-10%
- Maximum allowable slope for precise parking points (H/L): ≤ 1%

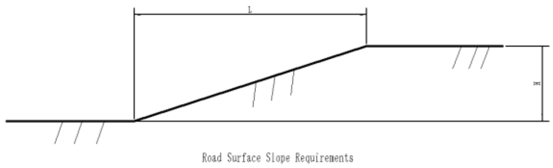

Fig 4.2: picture Road slope requirements

Step height

Maximum allowable step height: ≤ 5 mm

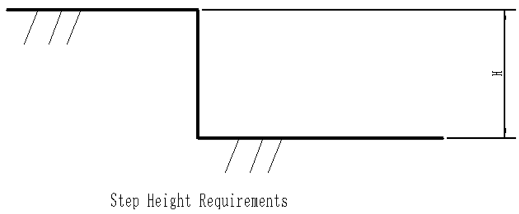

Fig 4.3: Step height requirements

Groove Width

Groove width refers to the maximum width of gaps or channels in the floor surface.

- Maximum allowable groove width: ≤ 8 mm
- If groove width exceeds this value, step height requirements apply

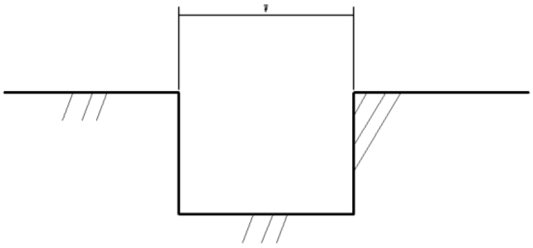

Fig 4.4: Groove width

Load-bearing requirements

- Static load calculation:

$$\text{Static load} = \frac{\text{Vehicle weight + Load}}{\text{Wheel system projected area}} \quad (kg/m^2)$$

- Dynamic load capacity must be ≥ 150% of static load
- Under maximum load, uneven ground settlement must be < 1/1000

Floor requirements

- Emery or hardened cement floors must be:
  - Wear-resistant
  - Dust-resistant
  - Anti-static
- Epoxy floors must receive anti-static treatment
- Waxing is prohibited
- Static friction coefficient (floor vs. PU wheels): ≥ 0.55
- Floor hardness: C30 or higher (≥ 30 MPa)

4.1.4 Personnel safety requirements

To ensure safe coexistence with personnel, the following rules apply:

- Personnel must actively avoid the AGV path and never intentionally obstruct or collide with it
- In narrow aisles (< 3.5 m), the AGV reduces speed and detection range; no sudden entry of people or obstacles permitted
- Keep routes clean and free of foreign objects (screws, hoses, cables, etc.)
- Do not modify drive mechanisms, control cabinets, or sensors
- Only trained and authorized operators may interact with the system
- No unauthorized electrical or software modifications
- Observe mechanical hazard warnings near moving parts
- Non-authorized persons must not access control panels during operation
- In emergencies, press any emergency stop switch and await qualified personnel
- Do not block LiDARs, cameras, or sensors

- The system is not waterproof or explosion-proof
- Perform visual inspection of mechanical/electrical components before startup
- Do not sit, strike, climb, lean on, or ride the AGV or load
- Cleaning permitted only when AGV is stopped and powered off

Minimum safety distance: Personnel should maintain at least 0.5 m clearance from the AGV envelope during motion (increased in narrow aisles per site risk assessment).

4.2 Aisle width requirements for each scenario (based on 1200*1200 pallet)

Aisle widths depend on load configuration, turning radius, speed, and single/dual-lane operation. Minimum values (based on vehicle kinematics and safety clearance) are:

Table 4.1: Aisle width Requirement

Scenario / Action	Channel Type	Minimum Width (m)
Straight travel (single lane)	Single lane only	1.8
Straight travel (dual lanes)	Dual lanes only	3.3
Turn (quarter turn)	Single channel right-angle turn	2.5
Turn (dual channel)	Dual-channel right-angle turn	3.3–4.0
Side pickup/unloading (single side)	Single channel side pickup	3.6
Side pickup/unloading (automatic both sides)	Single channel both sides	4.5
Dual-channel pickup/drop-off	Dual channels (one side traffic)	3.5
Dual-channel unloading both sides	Dual channels both sides	4.0 (4.2 for U-turn)
Dual-channel automatic pickup one side	Dual channels (one side pickup)	4.5
General dual-lane passage (recommended)	Dual lanes (pickup >3.5 m, main >3 m)	5.6 (for traffic control)

NOTE

- Values are minimum channel space (not marking space); actual site layout may require more for safety buffers or traffic rules.
- For dual-lane scenarios, main passage ≥ 3.5 m, unloading/pickup passage ≥ 3.0 m.
- Floor/ground markings must comply with ISO 3691-4 Annex A (Table A.1/A.2) for zone definition and personnel awareness (restricted, confined, hazard zones).

5 Wi-Fi communication

The XPL201P uses industrial-grade Wi-Fi for reliable two-way communication with the fleet management system (dispatch server). This connection enables real-time task management, status reporting, and safe operation across the warehouse

5.1 Wi-Fi Communication Module

The Wi-Fi module is the primary interface for data exchange between the AGV and the dispatch system. Its key functions include:

- Receiving structured task commands (pickup location, drop-off location, action type) from the fleet manager
- Transmitting real-time vehicle data: current position, operational status, battery level, alarms/warnings, and task progress
- Supporting dynamic rescheduling, fault reporting, and emergency interventions

The module operates on a stable industrial wireless network (2.4 GHz and 5 GHz bands) with high throughput and low latency. It ensures continuous connectivity within the defined operating area, allowing efficient task execution even in environments with moving obstacles or multiple vehicles.

Fig 5.1: Industrial Grade Wi-Fi Module

5.2 Wireless communication for automatic charging

The XPL201P supports automatic charging stations with wireless docking. The interaction follows a defined sequence coordinated by the fleet management server:

1. The server assigns a charging task and directs the AGV to a designated charging station.
2. Upon arrival, the AGV reports its position to the server.
3. The server instructs the charging station to initiate the charging process.
4. The charging station extends its telescopic charging arm, docks with the AGV's charging interface, and begins charging.
5. Continuous communication with FMS and Charger in order to monitor the status of charging.
6. When charging is complete (or interrupted), the server sends a stop-charging command.
7. The charging station retracts the telescopic arm to its fully retracted position.
8. The station confirms full retraction to the server (via feedback signal).
9. The server authorizes the AGV to depart safely.

6 Standard Pallet information

The XPL201P is designed to handle standard pallets, including Sichuan-shaped (three-stringer) and EPAL Euro pallets. Multiple pallet specifications are supported in the project, with the following key requirements and dimensions:

- Fork clearance requirement: ≤ 00 mm (minimum entry pocket height for fork insertion).
- Recommended gap between pallets: 200 mm (to allow safe AGV approach, alignment, and obstacle avoidance in storage areas). (Fig 6.1: Pallet Placement)

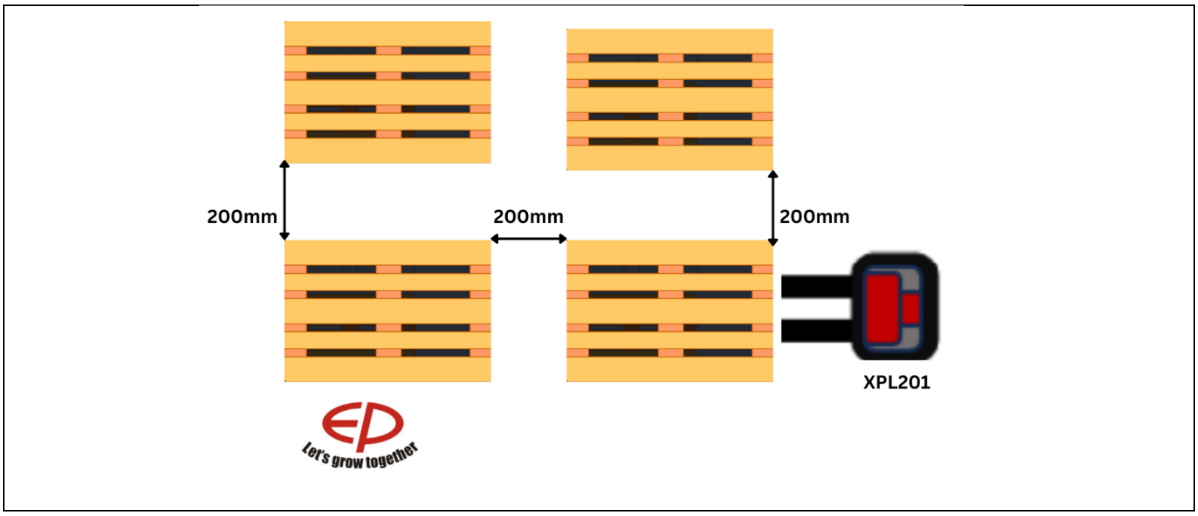

Fig 6.1: Pallet Placement

EPAL Euro pallet: Standard dimensions (as per EPAL specification):

- Length: 1200 mm
- Width: 800 mm
- Height: 144 mm
- Deck boards: 22 mm thick, 145 mm wide (top), 100 mm wide (bottom)
- Stringers: 145 mm high, 100 mm wide
- Blocks: 145 mm × 100 mm × 78 mm (corner), 145 mm × 145 mm × 78 mm (center)
- Fork entry: Four-way access, with clearance optimized for AGV forks.
- Refer to the illustration below for detailed EPAL pallet dimensions and structure.
- (Fig 6.2: EPAL Pallet Dimensions)

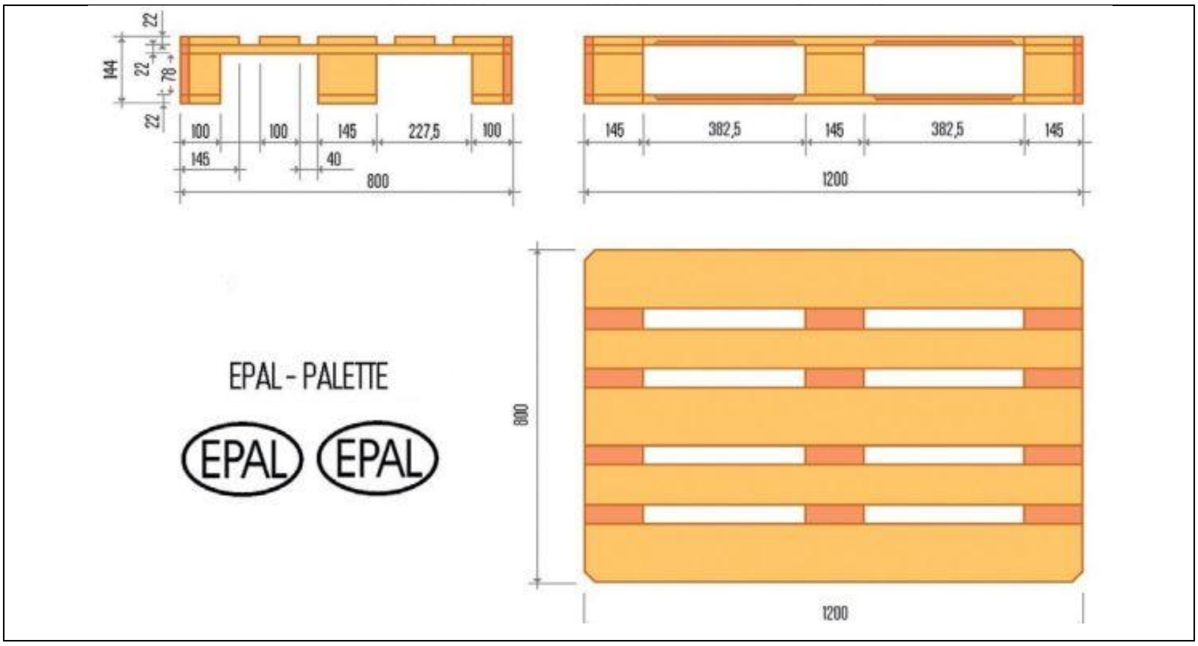

Fig 6.2: Pallet Dimensions